# UMC 0.11um LOGIC/MIXED_MODE Al Advanced Enhancement Vertical PNP BJT Monte Carlo SPICE Model Document

**Version 0.3 Phase 1**
**2011/08/19**

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# 1 Introduction

## 1.1 Revision History

**Table 1-1 Revision history**

| Version | Date | Author | Remark |
|---|---|---|---|
| 0.2_p1 | 2010/07/07 | Eason Cheng | Original release.<br>This Monte Carlo model including both process and mismatching characteristics are extracted from<br>L110AE_BJT_V011.mdl<br>L110AE_BJT_V011.lib |
| 0.3_p1 | 2011/08/19 | Eason Cheng | This Monte Carlo model including both process and mismatching characteristics are extracted from<br>l110ae_bjt_pnp_v041.mdl<br>l110ae_bjt_pnp_v041.lib |

## 1.2 General Information

This document serves as a reference to the design of products that will be manufactured by UMC using the following process.

> UMC 0.11 um Logic and Mixed-Mode 1P8M Metal Metal Capacitor Al Advanced Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

> G-02-LOGIC/MIXED_MODE11-1P8M-MMC/AL/L110AE-EDR

## 1.3 Model Usage

### 1.3.1 Simulators

The compact model is provided in various formats. It is guaranteed that the simulation results of this model from different simulators show differences less than 0.5% and 1%. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

Synopsys HSPICE release 2007.9
Cadence Spectre version 6.2.1.299
Mentor Graphics ELDO version 6.10_1.1

### 1.3.2 The combination of statistical model & corner model

The Monte-Carlo model and corner model (3 corners) are combined into a single model library. User can choose either the Monte-Carlo model or a corner case by selecting different model sections for simulation. The following sections are included:

MC: Monte-Carlo BJT model
TT: Typical BJT model
FF: Fast BJT model (Maximum beta model)
SS: Slow BJT model (Minimum beta model)

### 1.3.3 Mismatching parameters in corner model

The mismatching parameters are included in the corner models. Therefore, users can run Monte-Carlo mismatching simulation over a corner model. For example, users can evaluate the mismatching performance at the FF case.

### 1.3.4 Sigma number for Monte-Carlo model

Users can define standard deviation at sigma level for Monte-Carlo simulation. If a larger sigma number is used, it means the model parameters will have greater variation. Sigma=3 is recommended to be used.

### 1.3.5 Model usage

**The definitions of sub-circuit model parameters**

After invoking Monte-Carlo model parameters, the compact resistor models evolve to sub-circuit models. The input parameters for the sub-circuit model are the same as the compact model, except input parameter (mis_flag) are added for the sub-circuit model.

The following is the parameter list:

mis_flag: the flag parameter is used to turn on/off mismatching parameters for individual devices. 1 is turn-on and 0 is turn-off. Default is 1.

Follow these steps to invoke the model file during simulation.

-Define the flag of Monte-Carlo simulation.

PROCESS: 1 to turn on the MC process simulation
0 to turn off the MC process simulation

Example:

.param process=1
.param process=0

-Define sigma value for MC simulation.

Example:

.param sigma=3

-Define the number of Monte Carlo iterations.

Example:

sweep monte=500 (for HSPICE and ELDO)
montecarlo numruns=500 (for SPECTRE)

**For Synopsys HSPICE**

- Copy following files into your simulation directory:
  ModelFileName_MC.lib
  ModelFileName_MC.mdl
  ModelFileName_MC_CORNER.lib
  ModelFileName_MC_STATISTICAL_P.mdl
  ModelFileName_MC_STATISTICAL_M.mdl
- Add the following statement in the input file to the simulator:
  .lib "./ModelFileName_MC_CORNER.lib" CaseName
  , where CaseName could be MC, TT, FF, SS.
- Define device condition by 'Qxx' statement.
  Example:

  X1 Vcc Vbb Vee DeviceName

**For Cadence SPECTRE**

- Copy following files into your simulation directory:
  ModelFileName_MC.lib.scs
  ModelFileName_MC.mdl.scs
  ModelFileName_MC_CORNER.lib.scs
  ModelFileName_MC_STATISTICAL_P.mdl.scs
  ModelFileName_MC_STATISTICAL_M.mdl.scs

- Add the following statement in the input file to the simulator:
  include "./ModelFileName_MC_CORNER.lib.scs" section=CaseName
  , where CaseName could be mc, tt, ff, ss.
- Define device condition by 'Qxx' statement.

Example:

X1 (Vcc Vbb Vee) DeviceName

**For Mentor Graphics ELDO**

- Copy following files into your simulation directory:
  ModelFileName_MC.mdl.eldo
  ModelFileName_MC_STATISTICAL_P .mdl.eldo
  ModelFileName_MC.lib.eldo
  ModelFileName_MC_CORNER.lib.eldo
- Add the following statement in the input file to the simulator:
  .lib "./ModelFileName_MC_CORNERlib.eldo" CaseName
  , where CaseName could be MC, TT, FF, SS.
- Define device condition by 'Qxx' statement.

Example:

X1 Vcc Vbb Vee DeviceName

This model is valid within the following geometry and temperature ranges. Device behavior beyond these ranges may not be well described by the model and is not guaranteed.

**Geometry range:**

| DeviceName | Emitter Area | Base Area |
|---|---|---|
| PNP_V45X45_L110E | 4.5 x 4.5 $um^2$ | 10.98 x 10.98 $um^2$ |
| PNP_V90X90_L110E | 9.0 x 9.0 $um^2$ | 15.48 x 15.48 $um^2$ |
| PNP_V135X135_L110E | 13.5 x 13.5 $um^2$ | 19.98 x 19.98 $um^2$ |
| PNP_SV45X45_L110E | 4.5 x 4.5 $um^2$ | 10.98 x 10.98 $um^2$ |
| PNP_SV90X90_L110E | 9.0 x 9.0 $um^2$ | 15.48 x 15.48 $um^2$ |
| PNP_SV135X135_L110E | 13.5 x 13.5 $um^2$ | 19.98 x 19.98 $um^2$ |

**Temperature range:**

-50 °C ~ +125 °C

# 2 Monte Carlo DC Simulation Results for Process Variation

Five hundred (500) Monte Carlo iteration performed in the following simulation conditions, and the following corner points show TT, FF and SS values, individually

A reasonable number of Monte Carlo iteration is 30. The statistical significance of 30 iterations is quite high. If the circuit operates correctly for all 30 iterations, there is 99% probability that over 80% of all possible component values operate correctly. The relative error of a quantity determined through Monte Carlo analysis is proportional to (iteration times)-1/2.

## 2.1 Simulation Result Plots

### 2.1.1 PNP_V45X45_L110E

The following figures show the Monte Carlo simulation results of BJT's DC process variation at $V_{BE}$=-0.7V of PNP_V45X45_L11AE at 25 ℃:

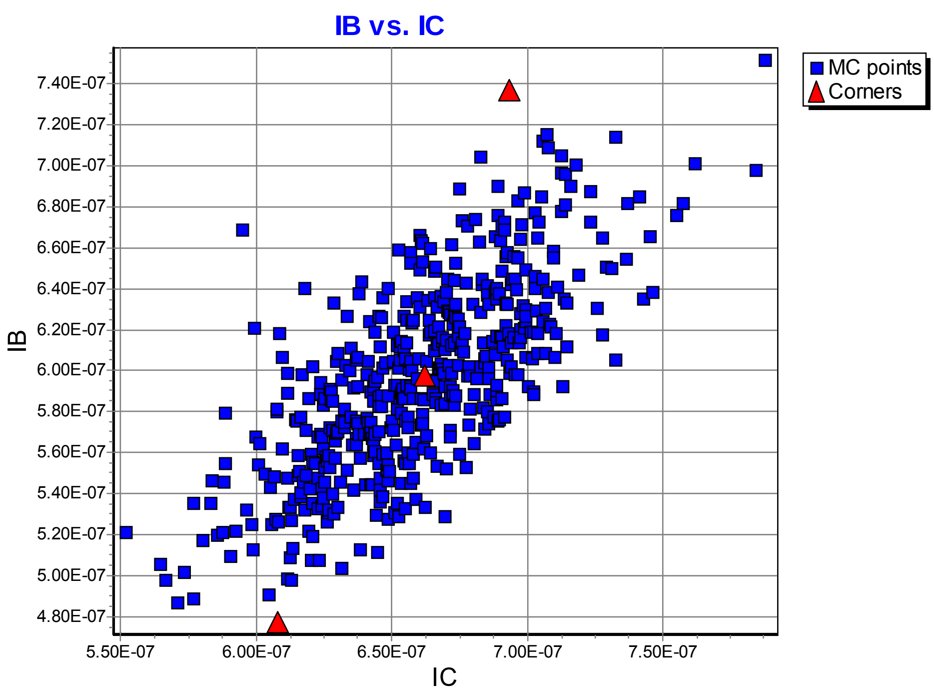


**Figure 2-1 Monte Carlo plots of “IB vs. IC of PNP_V45X45_L110E @ $V_{BE}$=-0.7V”**

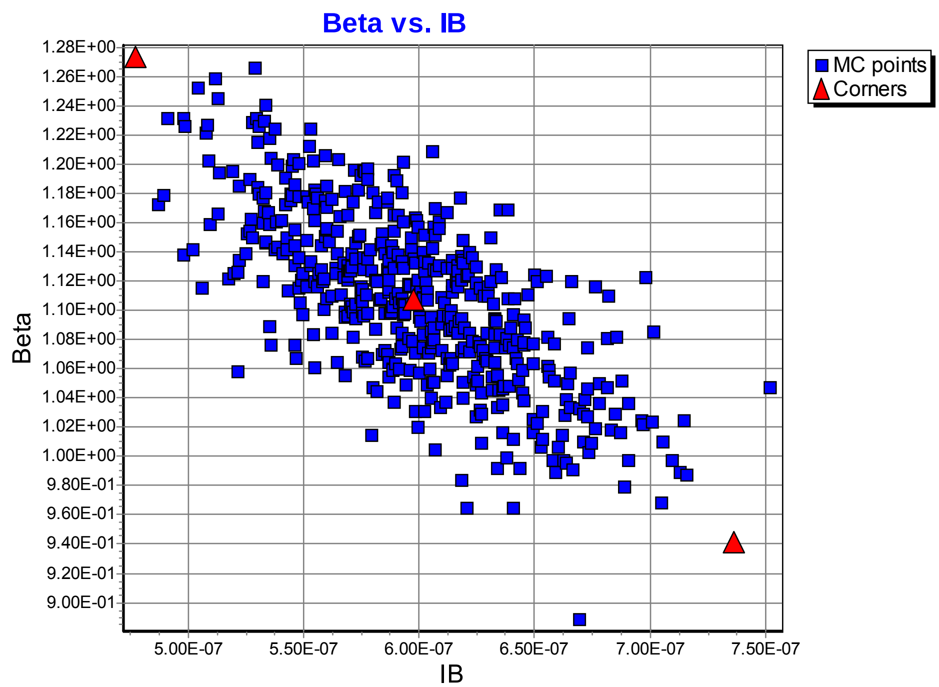


**Figure 2-2 Monte Carlo plots of "Beta vs. IB of PNP_V45X45_L110E @ $V_{BE}$=-0.7V"**

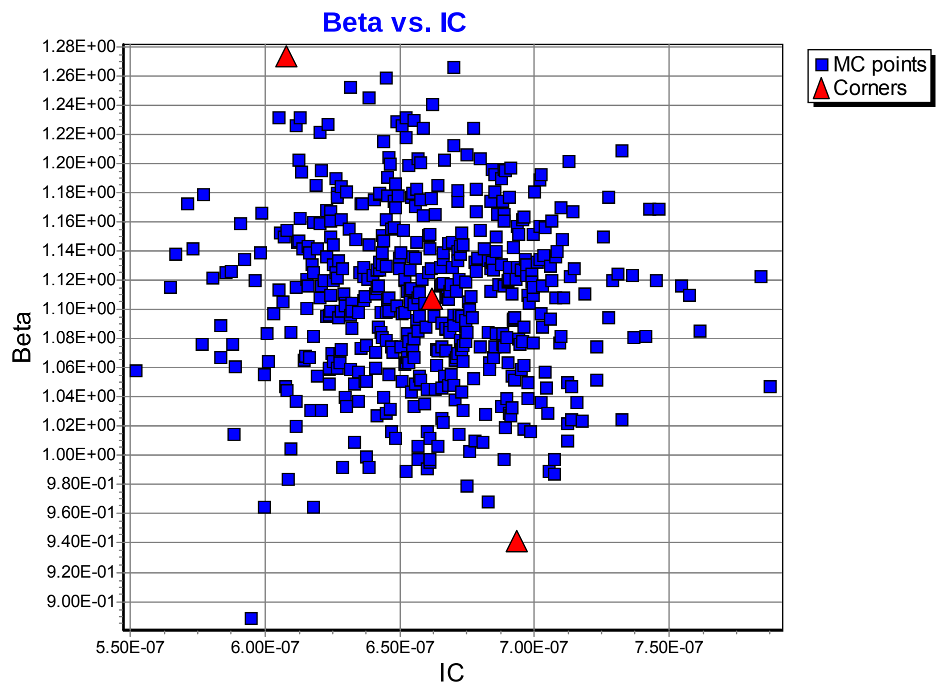


**Figure 2-3 Monte Carlo plots of "Beta vs. IC of PNP_V45X45_L110E @ $V_{BE}$=-0.7V"**

## 2.1.2 PNP_V90X90_L110E

The following figures show the Monte Carlo simulation results of BJT's DC process variation at $V_{BE}$=-0.7V of PNP_V90X90_L110E at 25 ℃:

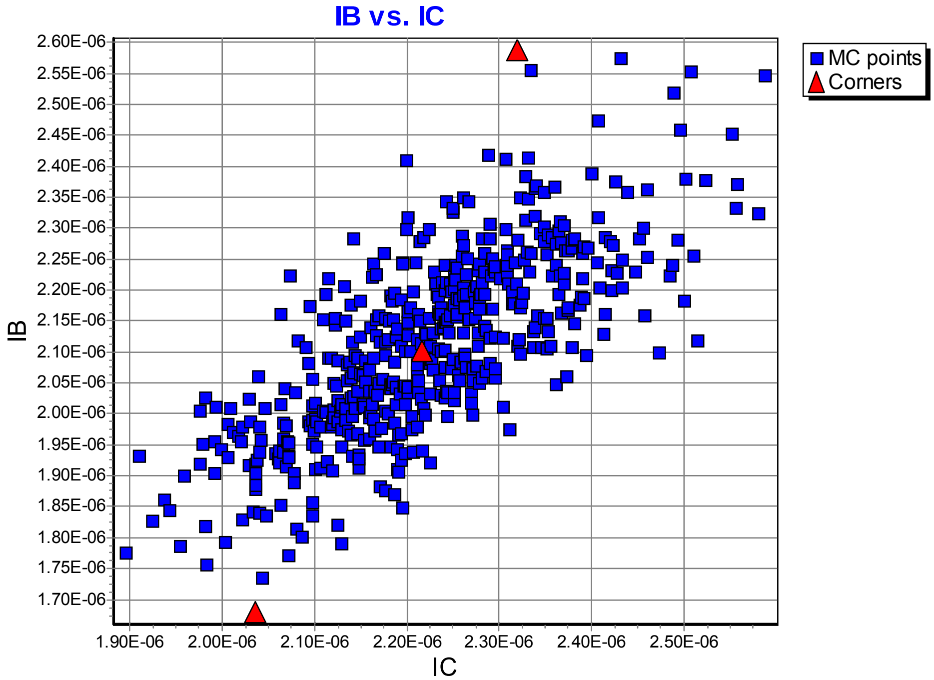


**Figure 2-4 Monte Carlo plots of "IB vs. IC of PNP_V90X90_L110E @ $V_{BE}$=-0.7V"**

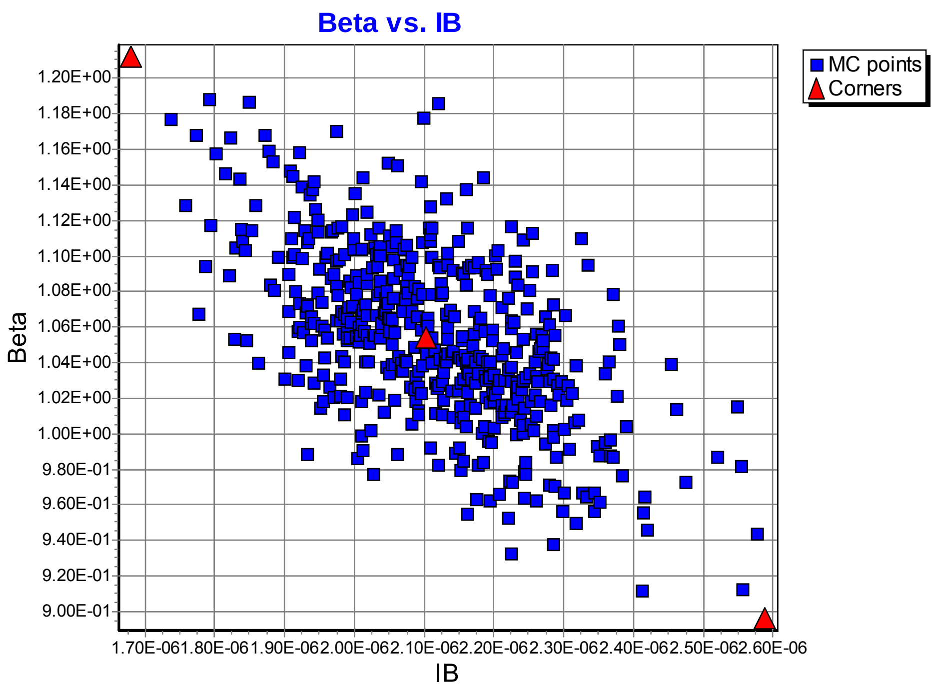

**Figure 2-5 Monte Carlo plots of "Beta vs. IB of PNP_V90X90_L110E @ $V_{BE}$=-0.7V"**

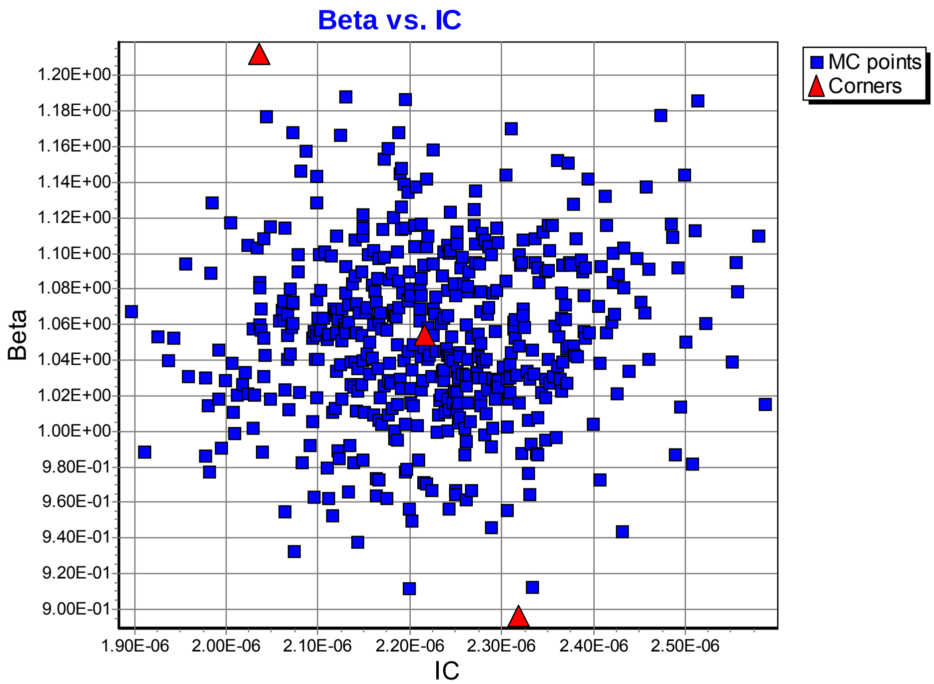


**Figure 2-6 Monte Carlo plots of "Beta vs. IC of PNP_V90X90_L110E @ $V_{BE}$=-0.7V"**

### 2.1.3 PNP_V135X135_L110E

The following figures show the Monte Carlo simulation results of BJT's DC process variation at $V_{BE}$=-0.7V of PNP_V135X135_L110E at 25 ℃:

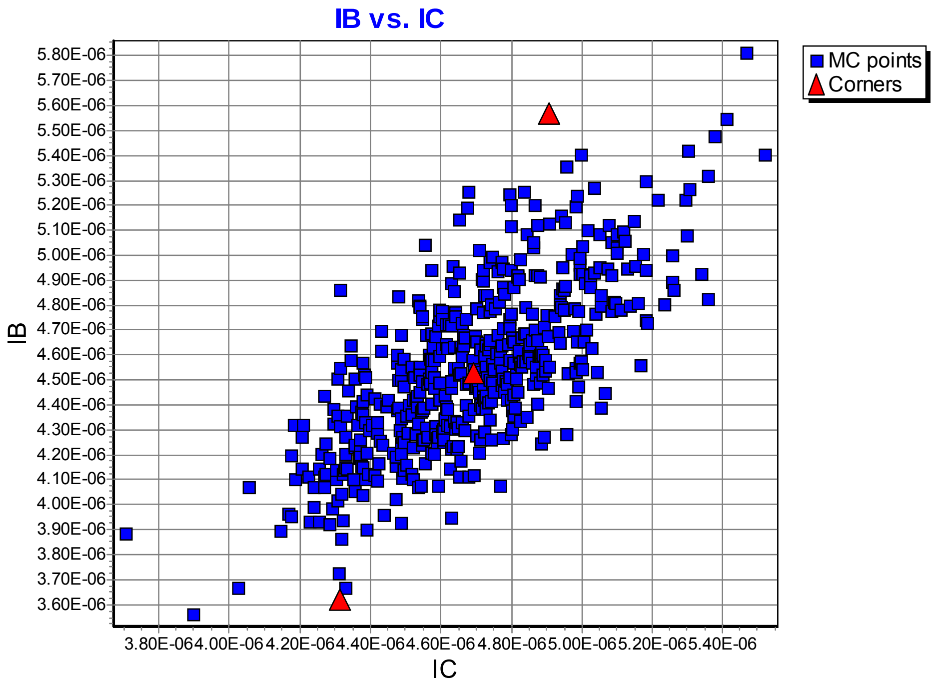


**Figure 2-7 Monte Carlo plots of "IB vs. IC of PNP_V135X135_L110E @ $V_{BE}$=-0.7V"**

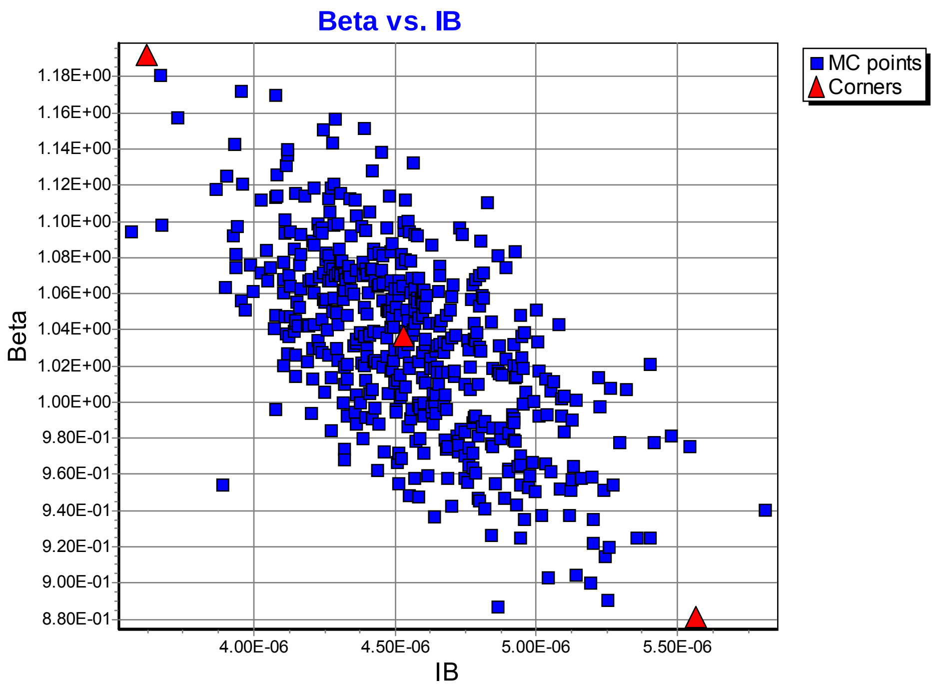


**Figure 2-8 Monte Carlo plots of "Beta vs. IB of PNP_V135X135_L110E @ $V_{BE}$=-0.7V"**

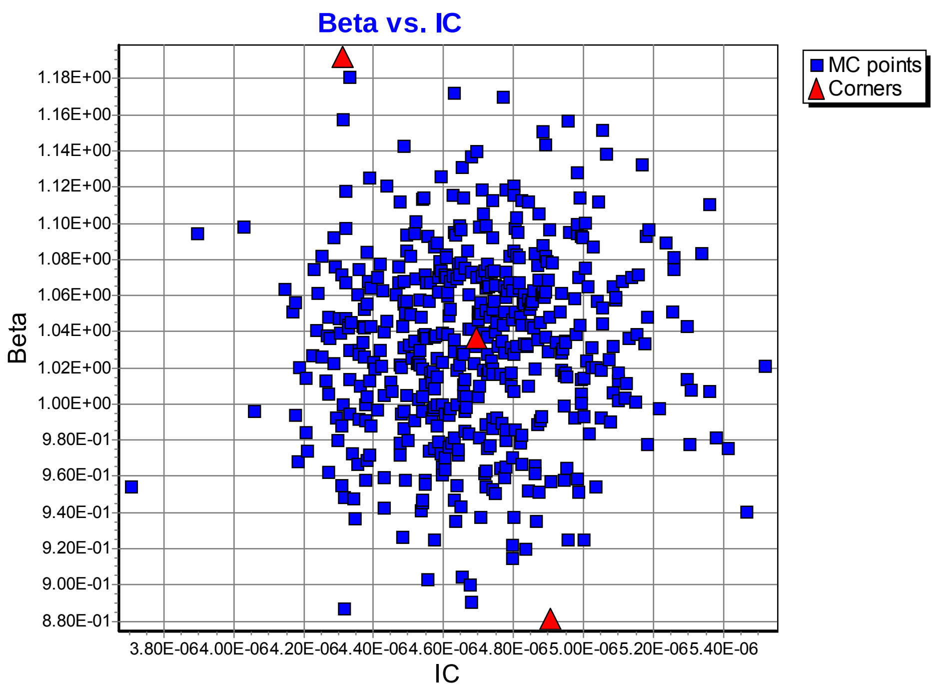


**Figure 2-9 Monte Carlo plots of "Beta vs. IC of PNP_V135X135_L110E @ $V_{BE}$=-0.7V"**

## 2.1.4 PNP_SV45X45_L110E

The following figures show the Monte Carlo simulation results of BJT's DC process variation at $V_{BE}$=-0.7V of PNP_SV45X45_L11E at 25 ℃:

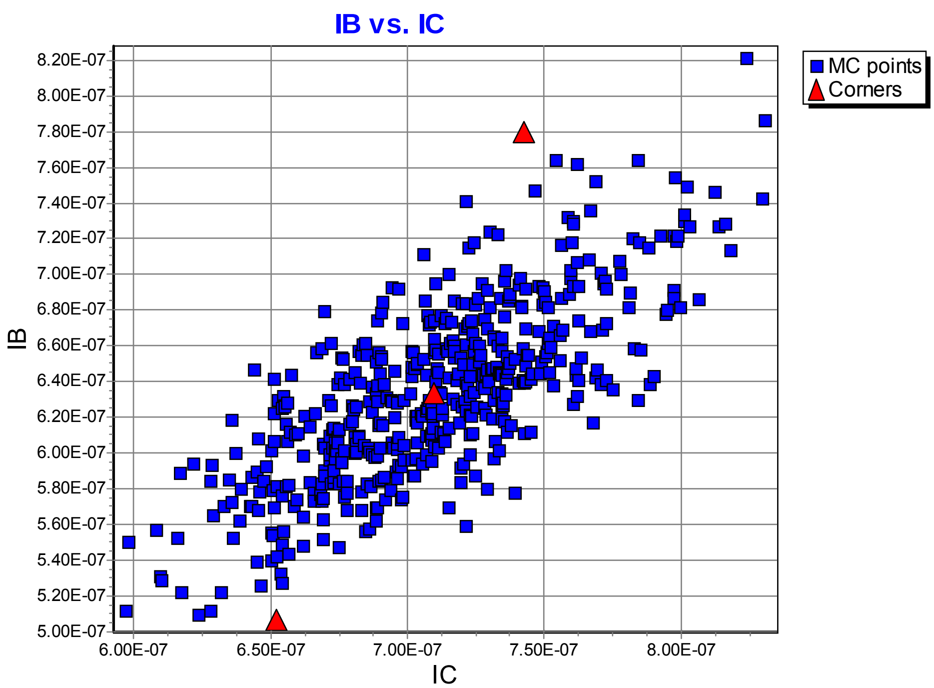


**Figure 2-10 Monte Carlo plots of "IB vs. IC of PNP_SV45X45_L110E @ $V_{BE}$=-0.7V"**

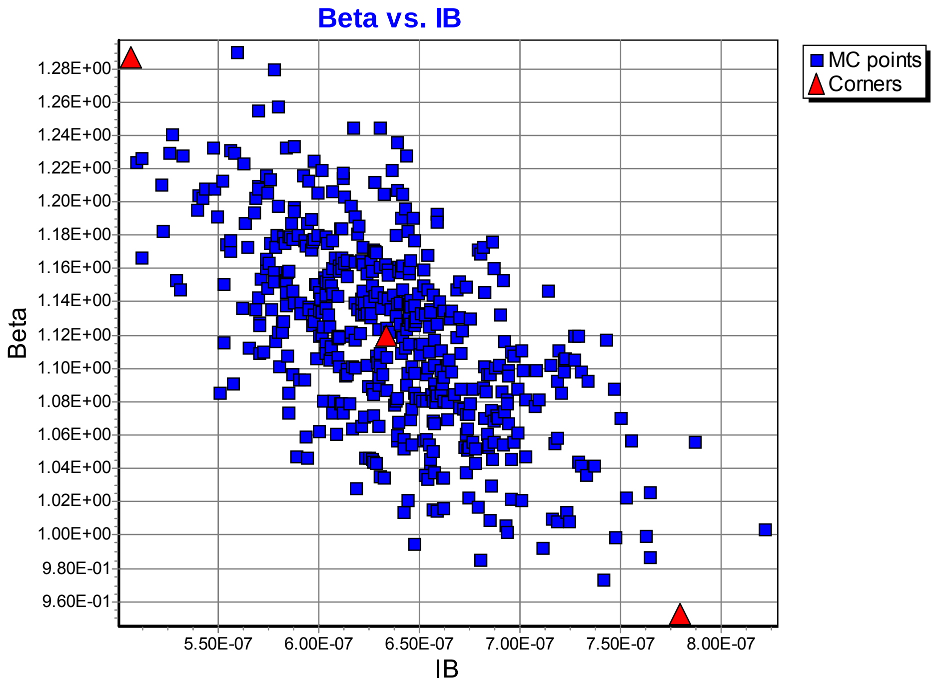


**Figure 2-11 Monte Carlo plots of "Beta vs. IB of PNP_SV45X45_L110E @ $V_{BE}$=-0.7V"**

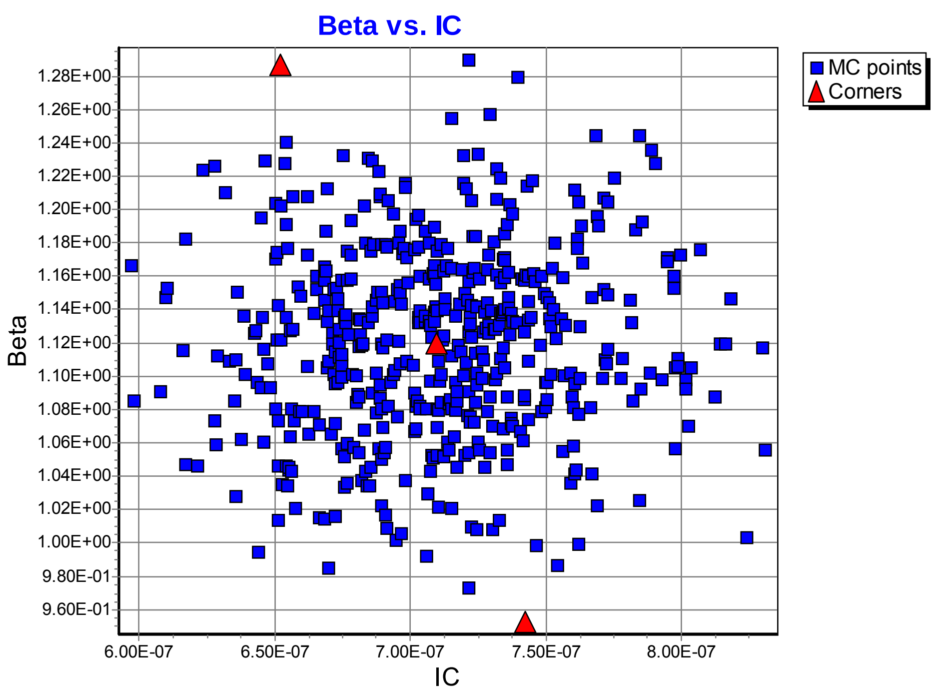


**Figure 2-12 Monte Carlo plots of "Beta vs. IC of PNP_SV45X45_L110E @ $V_{BE}$=-0.7V"**

### 2.1.5 PNP_SV90X90_L110E

The following figures show the Monte Carlo simulation results of BJT's DC process variation at $V_{BE}$=-0.7V of PNP_SV90X90_L110E at 25 ℃:

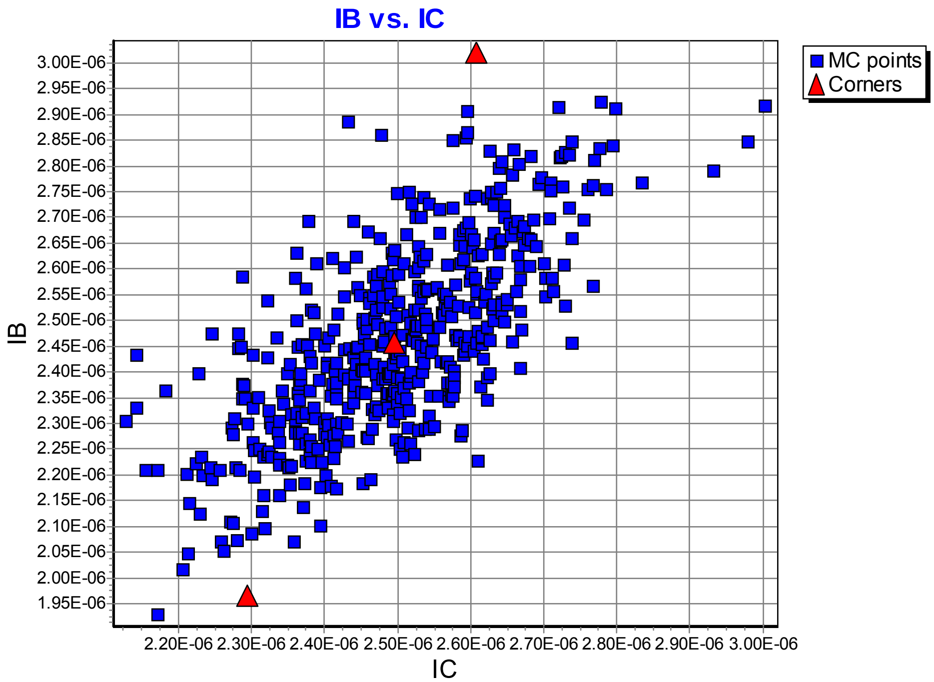


**Figure 2-13 Monte Carlo plots of "IB vs. IC of PNP_SV90X90_L110E @ $V_{BE}$=-0.7V"**

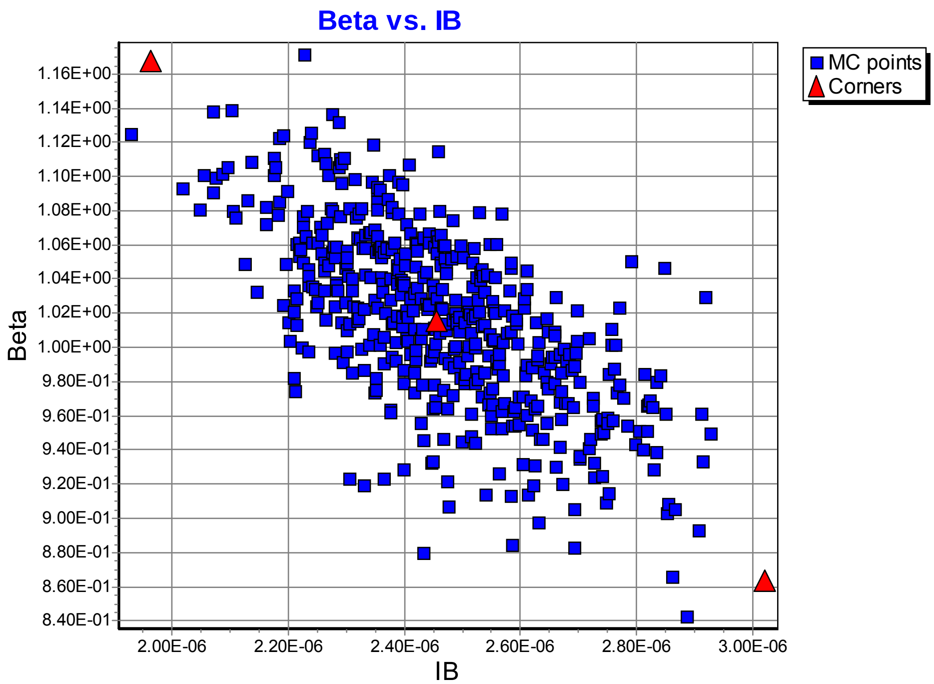

**Figure 2-14 Monte Carlo plots of "Beta vs. IB of PNP_SV90X90_L110E @ $V_{BE}$=-0.7V"**

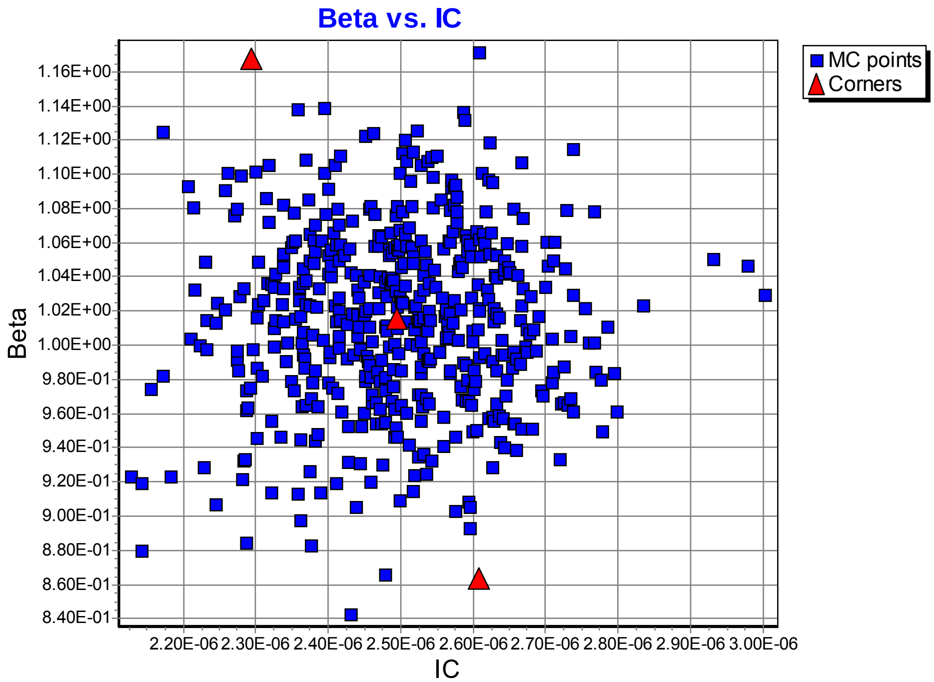


**Figure 2-15 Monte Carlo plots of "Beta vs. IC of PNP_SV90X90_L110E @ $V_{BE}$=-0.7V"**

### 2.1.6 PNP_SV135X135_L110E

The following figures show the Monte Carlo simulation results of BJT's DC process variation at $V_{BE}$=-0.7V of PNP_SV135X135_L110E at 25 ℃:

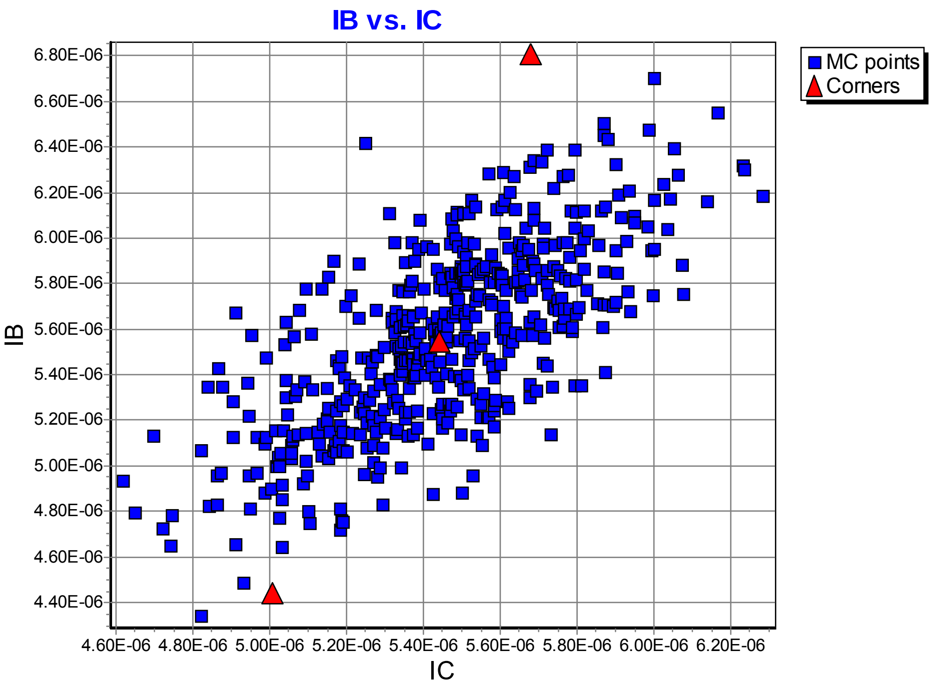


**Figure 2-16 Monte Carlo plots of "IB vs. IC of PNP_SV135X135_L110E @ $V_{BE}$=-0.7V"**

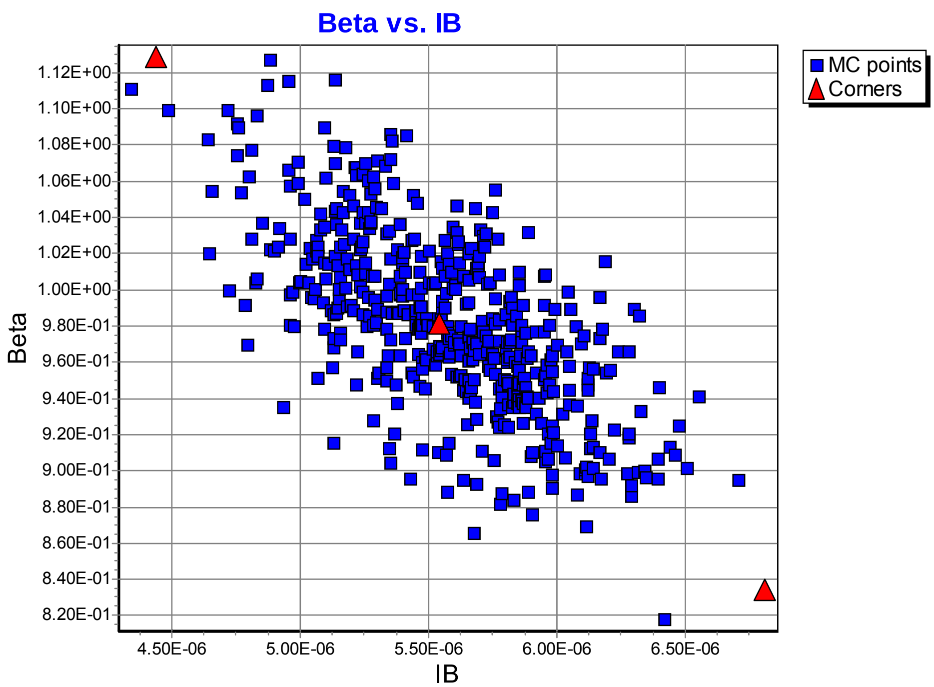


**Figure 2-17 Monte Carlo plots of "Beta vs. IB of PNP_SV135X135_L110E @ $V_{BE}$=-0.7V"**

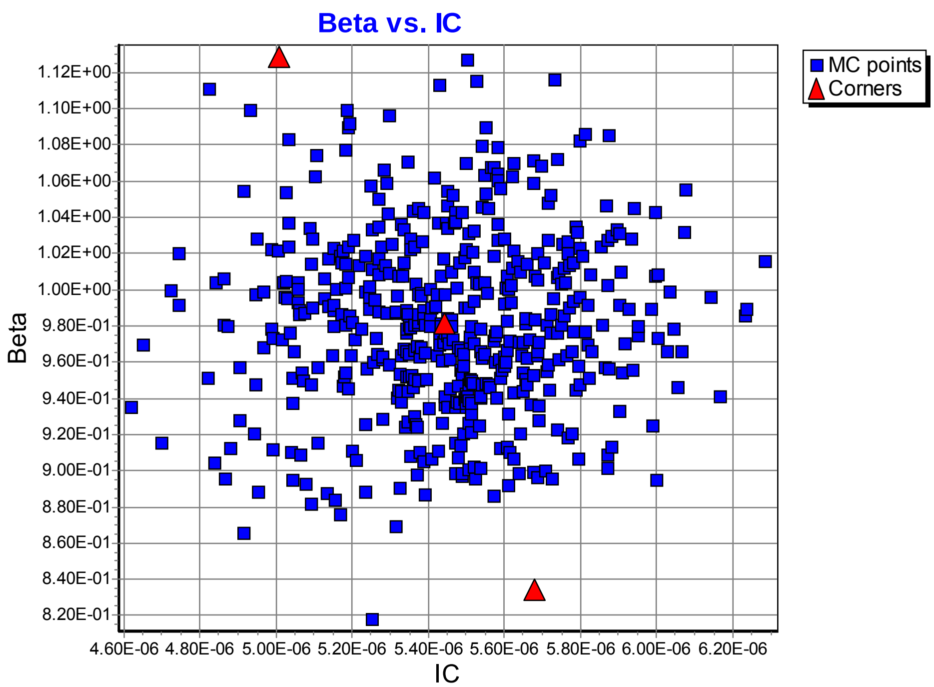


**Figure 2-18 Monte Carlo plots of "Beta vs. IC of PNP_SV135X135_L110E @ $V_{BE}$=-0.7V"**

# 3 Monte Carlo Simulation Results for Mismatch

## 3.1 Simulation Result Plots

The following figures show the Monte Carlo mismatching simulation results at 25 ℃.

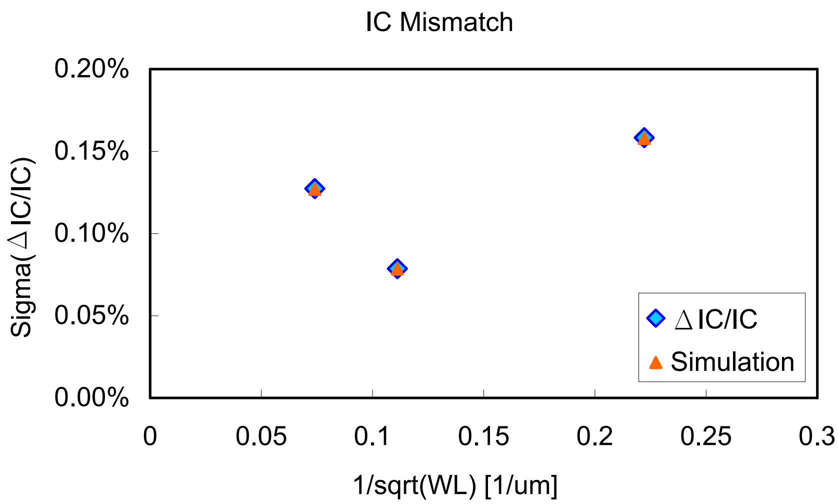


**Figure 3-1 Mismatching simulation of IC for PNP BJT device**

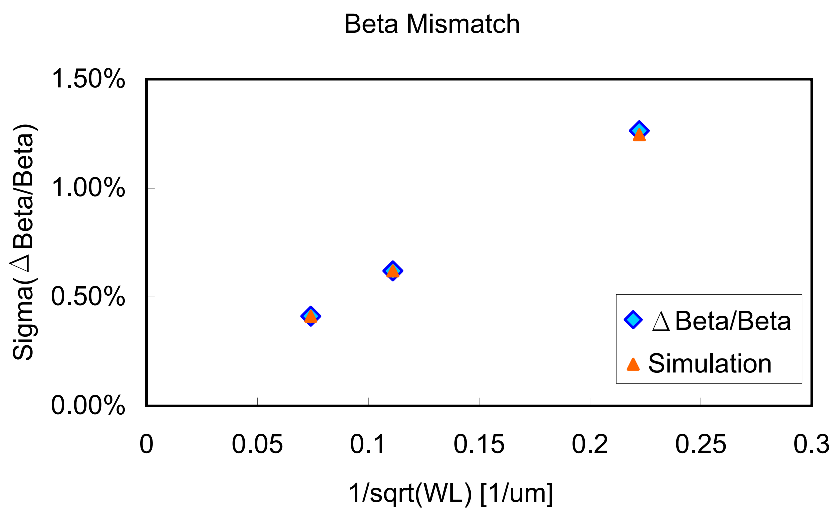


**Figure 3-2 Mismatching simulation of Beta for PNP BJT device**

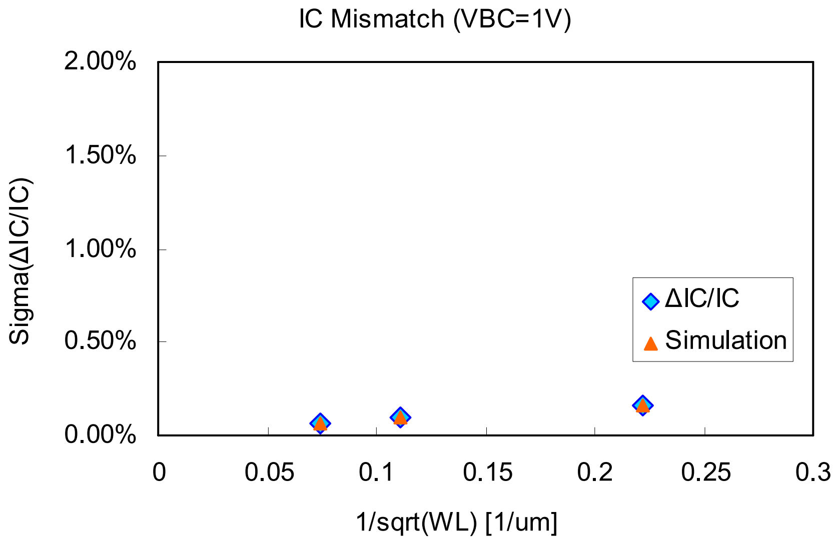


**Figure 3-3 Mismatching simulation of IC for PNP BJT with SAB device**

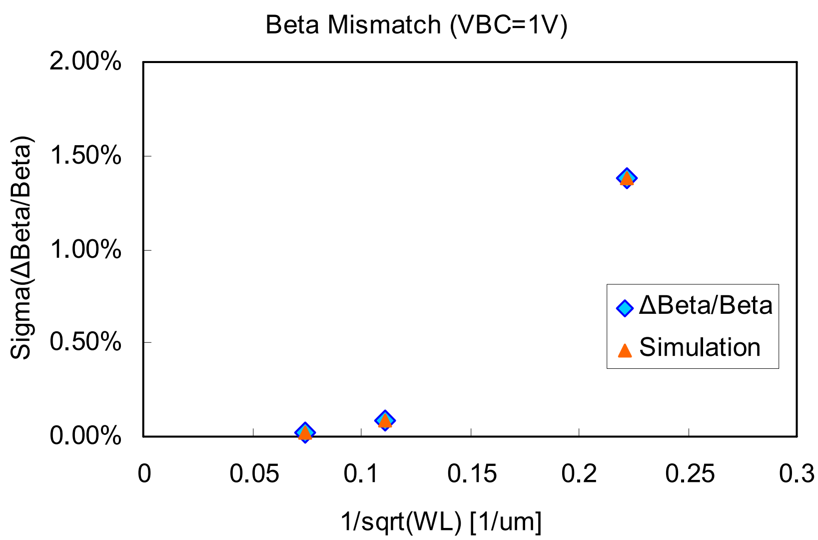


**Figure 3-4 Mismatching simulation of Beta for PNP BJT with SAB device**

# 4 Appendix

## 4.1 HSPICE Warning Message

**Table 4-1 Hspice Warning Message**

| NO. | Warning Message | Explanation |
|---|---|---|
| 1. | None | None |

## 4.2 ELDO Warning Message

**Table 4-2 Eldo Warning Message**

| NO. | Warning Message | Explanation |
|---|---|---|
| 1. | None | None |

## 4.3 SPECTRE Warning Message

**Table 4-3 Spectre Warning Message**

| NO. | Warning Message | Explanation |
|---|---|---|
| 1. | None | None |